

Ahmad Nouh

AI / ML Engineer | Applied AI, LLM Applications & NLP

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PROFESSIONAL SUMMARY

AI/ML Engineer with approximately six years of professional experience in Arabic natural language processing (NLP), large language model (LLM) applications, retrieval, and Python services. Delivered recommendation and classification services, built ML training and evaluation workflows, and cut an LLM extraction pipeline from about 15 minutes to under 3.

TECHNICAL SKILLS

ML and NLP: TensorFlow/Keras, PyTorch, scikit-learn, Hugging Face Transformers, PEFT/QLoRA, Arabic NLP, text classification, NER, OCR, pandas, NumPy.

LLMs and retrieval: LangGraph, LangChain, retrieval-augmented generation (RAG), tool calling, MCP, Azure OpenAI, Gemini, embeddings, hybrid search, PostgreSQL/pgvector, Neo4j.

Backend: Python, SQL, FastAPI, Flask, REST APIs, Celery, RabbitMQ, Redis; Go, Java, Node.js, TypeScript.

MLOps and cloud: MLflow, DVC, model evaluation, Docker, Git, CI/CD, Azure Machine Learning, AWS EC2 GPUs, Grafana.

PROFESSIONAL EXPERIENCE

Blink

Nov 2025 - Jul 2026

AI Engineer

Remote | Company HQ: Falls Church, Virginia, US

- Delivered attendee and session recommendations using Python/FastAPI, PostgreSQL/pgvector, and RabbitMQ, with automated tests, CI/CD, and staged releases.
- Designed and evaluated a beta, feature-flagged LangGraph flight-creation workflow with five tools for booking, attendee search, location lookup, and document parsing.
- Built an optical character recognition (OCR)-assisted workflow with Docling and Gemini 2.5 Flash to extract structured flight-ticket and hotel data from PDFs and images.
- Benchmarked Gemma 3 12B with Ollama on AWS EC2 G5 GPUs to assess self-hosting feasibility, measuring time to first token, throughput, latency, memory, and GPU utilization.

MENT

Oct 2023 - Oct 2025

AI Engineer

Remote | Company HQ: Falls Church, Virginia, US

- Cut AI pipeline runtime from **about 15 minutes to under 3 minutes** by parallelizing asynchronous LLM extraction calls and moving schema mapping and validation into Python.
- Improved data-to-profile attribution by approximately 33% versus the prior system, reducing ambiguity for people with identical or similar names.
- Contributed to multi-agent enrichment workflows in n8n, including MCP tool integration, RabbitMQ integration and retries, and Grafana tracing.
- Designed and implemented a Neo4j relationship knowledge graph for shared-interest, employer, and multi-hop network insights.
- Built Python microservices and Celery workers for enrichment workloads, and integrated Azure OpenAI into retrieval and conversational services.

PROFESSIONAL EXPERIENCE CONTINUED

Lableb

Aug 2020 - Sep 2023

AI Engineer

Remote | Company HQ: Dubai, UAE

- Built a production B2B spam-classification service handling **up to 30,000 requests per day** and filtering **about 90% of spam queries**.
- Compared n-gram and neural noisy-channel approaches for Arabic spelling correction; adopted KenLM to improve runtime while retaining comparable results.
- Trained and evaluated Arabic NER, POS tagging, transliteration, and spelling-correction models using TensorFlow/Keras and Azure Machine Learning; tracked F1, precision, recall, and accuracy, and versioned datasets and model artifacts with DVC.
- Built NLP microservices using Python, Go, Flask, Spring Boot, and Go Fiber.

AI Engineer Intern | Lableb | Apr 2020 - Jul 2020

Code Experts

Oct 2018 - Jan 2020

Web Developer

Damascus, Syria | On-site

- Built Angular and Ionic interfaces for video-on-demand and taxi-booking products; contributed Node.js backend work and a WebSocket chat feature connecting drivers and riders.

SELECTED PROJECTS

Second Memory | Personal knowledge assistant

Built a self-hosted knowledge assistant with FastAPI, LangGraph, PostgreSQL/pgvector, and Flutter. Fused vector and full-text search over saved summaries with Reciprocal Rank Fusion (RRF); added account-scoped filters, keyword fallback, streamed answers, validated references, CI tests, and a labeled Recall@k/MRR evaluation runner.

SILA-4B | Arabic LLM fine-tuning and evaluation

Built a Qwen3-4B Arabic tool-calling experiment with PyTorch, Transformers, and PEFT/QLoRA, using 4-bit NF4 on an 8 GB GPU. Prepared 6,600 synthetic examples (5,280 train / 1,320 validation) with leakage checks; implemented resumable base-versus-adaptor evaluation on 100 Arabic and 500 English BFCL cases.

End-to-End Topic Classification

Built an MLflow pipeline connecting training, hyperparameter tuning, prediction, experiment artifacts, and class/slice evaluation. Used Snorkel to inspect emoji and short-tweet slices; added Streamlit views for exploring data, evaluating behavior, and running inference.

EDUCATION

Bachelor's degree in Information Technology Engineering

Damascus University | 2015 - 2021 | Artificial Intelligence specialization

LANGUAGES

Arabic: Native or bilingual | English: Professional working | Turkish: Elementary

SELECTED CERTIFICATIONS

DeepLearning.AI TensorFlow Developer | July 2021

Introduction to Machine Learning in Production | DeepLearning.AI